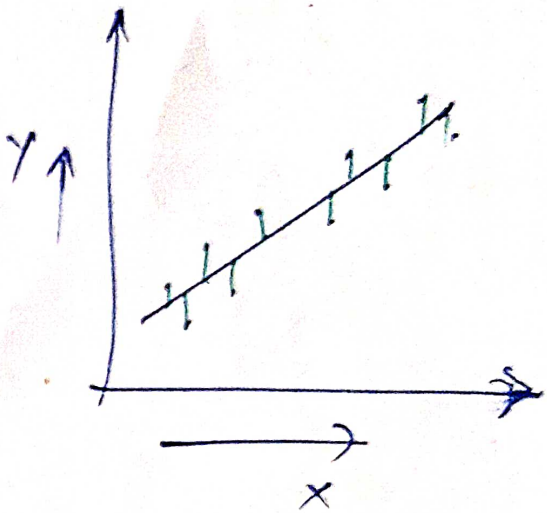
Linear Regression Derivation

- Get start with N inputs and outputs.—
  (x_i, y_i)

- Define the line of the best fit line as—

$$\hat{Y}_i = a + Bx_i$$



- Minimize the cost f^n.

Squared errors ⇒

$$S = \sum_{i=1}^{n} (Y_i - \hat{Y}_i)^2$$

Hence,

$$S = \sum_{i=1}^{n} (Y_i - a - Bx_i)^2$$

To minimize. do the first order derivative with respect to 'a' and 'B'

$$\frac{\partial S}{\partial a} \left[\sum_{i=1}^{n} (Y_i - a - Bx_i)^2 \right]$$

it should be equal to zero.

$$\therefore \frac{\partial S}{\partial a}\left[\sum_{i=1}^{n}\left(Y_i - a - Bx_i\right)^2\right] = 0$$

or. $$\sum_{i=1}^{n} -2\left(Y_i - a - Bx_i\right) = 0$$

or. $$\sum_{i=1}^{n}\left(Y_i - a - Bx_i\right) = 0$$

or. $$\sum_{i=1}^{n} Y_i - \sum_{i=1}^{n} a - B\sum_{i=1}^{n} x_i = 0$$

or $$\therefore \sum_{i=1}^{n} Y_i - na - B\sum_{i=1}^{n} x_i = 0$$

or. $$a = \frac{\sum_{i=1}^{n} Y_i - B\sum_{i=1}^{n} x_i}{n}$$

or. $$a = \frac{1}{n}\sum_{i=1}^{n} Y_i - B.\frac{1}{n}\sum_{i=1}^{n} x_i$$

or. $$\boxed{a = \bar{Y} - B\bar{x}}$$

$$\therefore a = \frac{1}{n}\sum_{i=1}^{n} Y_i - B\frac{1}{n}\sum_{i=1}^{n} x_i$$

$$\boxed{a = \frac{1}{n}\left(\sum_{i=1}^{n} Y_i - B\sum_{i=1}^{n} x_i\right)}$$

Finding out β.

Do the derivative of S with respect to β.

$$\frac{\partial S}{\partial \beta}\left[\sum_{i=1}^{n}\left(Y_i - a - \beta x_i\right)^2\right] = 0$$

or, $$\sum_{i=1}^{n} -2x_i\left(Y_i - a - \beta x_i\right) = 0$$

or, $$\sum_{i=1}^{n} x_i\left(Y_i - a - \beta x_i\right) = 0$$

or, $$\sum_{i=1}^{n}\left(x_i Y_i - a x_i - \beta x_i^2\right) = 0$$

$(a = \bar{Y} - \beta\bar{X})$

or, $$\sum_{i=1}^{n}\left(x_i Y_i - (\bar{Y} - \beta\bar{X})x_i - \beta x_i^2\right) = 0$$

or $$\sum_{i=1}^{n}\left(x_i Y_i - \bar{Y}x_i + \beta\bar{x}x_i - \beta x_i^2\right) = 0$$

or, $$\sum_{i=1}^{n}\left(x_i Y_i - \bar{Y}x_i\right) + \sum_{i=1}^{n}\left(\beta\bar{x}x_i - \beta x_i^2\right) = 0$$

or, $$\sum_{i=1}^{n}\left(x_i Y_i - \bar{Y}x_i\right) - \beta\sum_{i=1}^{n}\left(x_i^2 - \bar{x}x_i\right) = 0$$

or, $$\beta = \frac{\sum_{i=1}^{n}\left(x_i Y_i - \bar{Y}x_i\right)}{\sum_{i=1}^{n}\left(x_i^2 - \bar{x}x_i\right)}$$

Finding out B.

Do the derivative of S with respect to B.

$$\frac{\partial S}{\partial B}\left[\sum_{i=1}^{n}\left(Y_i - a - Bx_i\right)^2\right] = 0$$

or, $$\sum_{i=1}^{n} -2x_i\left(Y_i - a - Bx_i\right) = 0$$

or, $$\sum_{i=1}^{n} x_i\left(Y_i - a - Bx_i\right) = 0$$

or, $$\sum_{i=1}^{n}\left(x_iY_i - ax_i - Bx_i^2\right) = 0$$

$(a = \bar{Y} - B\bar{X})$

or, $$\sum_{i=1}^{n}\left(x_iY_i - (\bar{Y} - B\bar{X})x_i - Bx_i^2\right) = 0$$

or $$\sum_{i=1}^{n}\left(x_iY_i - \bar{Y}x_i + B\bar{x}x_i - Bx_i^2\right) = 0$$

or, $$\sum_{i=1}^{n}\left(x_iY_i - \bar{Y}x_i\right) + \sum_{i=1}^{n}\left(B\bar{x}x_i - Bx_i^2\right) = 0$$

or, $$\sum_{i=1}^{n}\left(x_iY_i - \bar{Y}x_i\right) - B\sum_{i=1}^{n}\left(x_i^2 - \bar{x}x_i\right) = 0$$

or, $$B = \frac{\sum_{i=1}^{n}\left(x_iY_i - \bar{Y}x_i\right)}{\sum_{i=1}^{n}\left(x_i^2 - \bar{x}x_i\right)}$$

or,

$$B = \frac{\sum_{i=1}^{n}(x_i y_i - \bar{y}\, x_i)}{\sum_{i=1}^{n}(x_i^2 - \bar{x}\, x_i)}$$

or,

$$B = \frac{\sum_{i=1}^{n} x_i y_i - \sum_{i=1}^{n} x_i \frac{1}{n}\sum_{i=1}^{n} y_i}{\sum_{i=1}^{n} x_i^2 - \frac{1}{n}\sum_{i=1}^{n} x_i \sum_{i=1}^{n} x_i}$$

or,

$$\boxed{B = \frac{n\sum_{i=1}^{n} x_i y_i - \sum_{i=1}^{n} x_i \sum_{i=1}^{n} y_i}{n\sum_{i=1}^{n} x_i^2 - \left(\sum_{i=1}^{n} x_i\right)^2}}$$